

Off-Grid SMS Agent: README

Off-Grid SMS Agent

Overview

The Off-Grid SMS Agent is a specialized, low-bandwidth AI backend service designed to operate over Google Voice (via Gmail IMAP/SMTP). It acts as a bridge between a user with limited or "off-grid" connectivity and powerful local AI models running on a host server.

By stripping away modern web bloat and using plain text SMS/MMS as the primary interface, this agent enables advanced capabilities--like weather radar analysis, offline Wikipedia research, plant identification, and video recipe extraction--on simple cellular connections where traditional apps would fail.

"Dual Brain" Architecture

The system uses a "Dual Brain" approach to balance speed and intelligence:

- 1. **Fast Brain (Lemonade)**: Handles logic, routing, and quick text responses.
 - Engine**: Lemonade (Llama CPP wrapper)
 - Model**: efficient logic models (e.g., Llama-3.2-1B, Qwen2.5-3B)
 - Role**: Chat, decision making, tool calling.
- 2. **Vision Brain (LM Studio)**: Handles image analysis and complex reasoning.
 - Engine**: LM Studio (OpenAI-compatible server)
 - Model**: Vision-Language Models (e.g., Qwen2.5-VL-7B, Llama-3.2-Vision)
 - Role**: Analyzing MMS photos (Plant ID), watching videos (!recipe), and detailed reasoning.

Running the System

To start the agent, you must ensure the "Dual Brain" backends are running, then launch the main script.

1. Start the Fast Brain (Lemonade)

This provides the core text intelligence.

- Open a terminal.
- Run the Lemonade server on port **8000**.

```
python lemonade_server.py --port 8000 --model models/Llama-3.2-1B-Instruct.gguf
```

2. Start the Vision Brain (LM Studio)

This enables Plant ID and Recipe features.

- Open **LM Studio**.
- Load your Vision Model (e.g., `Qwen2.5-VL-7B-Instruct`).
- Go to the **Developer/Server** tab.
- Start the Local Server on port **1234**.
- Ensure "Cross-Origin-Resource-Sharing (CORS)" is ON.

3. Start the Off-Grid Agent

- Double-click `run\_sms\_agent.bat` OR run:

```
python off_grid_automation/main.py
```

- The agent will initialize, check connections to both brains, and begin monitoring for SMS.

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Features & Services

1. Communication Core

- **Google Voice Bridge**: Uses Gmail's IMAP to read incoming SMS notifications and SMTP to reply.
- **Smart Rate Limiting**: Prevents carrier blocking by queuing messages (max 4 bursts, then delays).
- **Duplicate Detection**: Ignores identical messages sent within 60 seconds (prevents loops).

2. Weather & Radar Service

Direct integration with NOAA and OpenMeteo for hyper-local data.

- **!weather <loc>**: Returns current conditions and a concise 12-hour forecast.
- **!forecast <loc>**: Detailed hourly breakdown.
 - **Targeted Modes**: Filter by variable to save SMS space.
 - `!forecast(temp)(24) Seattle`` (Temp only, next 24h)
 - `!forecast(wind)(12) Seattle`` (Wind speed/dir)
 - `!forecast(precip)(48) Seattle`` (Rain prob/amount)
- **!temp-history <loc>**: Retrieves raw station data for the past 24h to show actual temperature trends.
- **!radar <loc> <miles>**: Generates a custom NEXRAD radar map.
 - **Static**: Sends a high-contrast PNG image.
 - **Animation**: Generates a GIF loop of the last 30 or 60 minutes.
 - Usage: `!radar Seattle 30 gif 30`` (30-min loop)
 - Usage: `!radar Seattle 50 gif 60`` (1-hour loop)

3. Vision Services

- **Plant Identification**:
 - Simply send a **Photo (MMS)** to the agent.
 - The agent detects the image and routes it to the Vision Brain.
 - Identifies species, edible/toxic lookalikes, and medicinal uses.
- **Recipe Extraction (!recipe)**:
 - Command: `!recipe <video_url>`
 - The agent downloads the video (YouTube/Instagram/TikTok).
 - Extracts frames and uses Vision AI to reconstruct the recipe (Ingredients + Steps).

4. Knowledge Engine (Wiki Agent)

A fully offline research tool capability.

- **Offline Archive**: Accesses a local ZIM file (e.g., all of Wikipedia or StackOverflow).
- **Search Protocol**:
 - The AI searches the archive for terms.
 - Reads article chunks.
 - Synthesizes an answer based *only* on the offline text.

5. System Utilities

- **!check**: Runs a full diagnostic of GPU, Disk, Connectivity, and AI Backends.
- **!clear**: Wipes your conversation history from the local database (privacy).
- **!test**: Simple ping to verify the system is awake.

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Command Reference

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COMMAND	SYNTAX	DESCRIPTION
Info	<code>`!info`</code>	Shows this command list + link to the PDF Manual.
Weather	<code>`!weather <loc>`</code>	Current weather + 12h Summary.
Radar	<code>`!radar <loc> <mi> [gif]`</code>	Weather radar map (Static or GIF).
Forecast	<code>`!forecast <loc>`</code>	Hourly forecast details.
History	<code>`!temp-history <loc>`</code>	Past 24h temperature observations.
Recipe	<code>`!recipe <url>`</code>	Extract recipe from video link.
Check	<code>`!check`</code>	System status diagnostics.
Clear	<code>`!clear`</code>	Delete conversation history.
Test	<code>`!test`</code>	Ping / Connectivity test.

> ****Full Manual****: A comprehensive PDF manual (``Off-Grid_Agent_Manual_Full.pdf``) is available and can be retrieved via the ``!info`` command.